

Ido Nitzan Hidekel

Ph.D. Student, Tel Aviv University • AI Research Scientist Intern, IBM Research



E-mail



Website



GitHub



Scholar



LinkedIn

Ph.D. student in Electrical Engineering at Tel Aviv University and **AI Research Scientist Intern** in the Multimodal AI group at **IBM Research**. Research on the emergence of **structural low rank and spectral bias** in deep learning and how they shape **generalization, forgetting, and optimization**, with applications to classification, language and vision models, INRs and graphics, forgetting and model merging, and post-training. Experienced in taking AI systems from **research through data engineering, model development, evaluation, and deployment**. First author of papers accepted to **ICML 2026** and the **ICML 2026 CATS Workshop**.

Experience

AI Research Scientist Intern — IBM Research

Oct 2026–present

Multimodal AI group, led by Eli Schwartz.

Graduate Researcher — Geometric Deep Learning Lab, Tel Aviv University

2024–present

Advisor: Prof. Dan Raviv

- Research how structural low rank and spectral bias emerge in deep networks and shape generalization, forgetting, and optimization.
- Develop theoretical predictors and reproducible empirical pipelines across vision, language, and audio models.

Lead, Deep Learning — Dtect Vision

Tel Aviv | 2025–2026

Led end-to-end algorithm development for large-scale synthetic-media detection (audio, video, and images).

- Developed and evaluated deep-learning systems using vision foundation models, including architecture design, fine-tuning, calibration, and out-of-distribution evaluation.
- Built a **5M+ sample data pipeline** with leakage audits, source-aware sampling, and stratified evaluation.
- Implemented distributed experimentation using **PyTorch DDP, GCP, Vertex AI, Docker, and W&B**.

Selected Publications

SONAR: Spectral-Contrastive Audio Residuals for Generalizable Deepfake Detection

Ido Nitzan Hidekel, Gal Lifshitz, Khen Cohen, Dan Raviv

ICML 2026, regular track. [Paper](#) | [Code](#) | [Project](#)

Developed an XLSR-based dual-path detector combining content and learnable residual representations. Achieved 1.45% EER on ASVspoof 2021 DF and 5.43% EER on In-the-Wild, with robustness to codec and bandwidth shifts.

Catastrophic Forgetting is Low-Rank: A Function-Space Theory for Continual Adaptation

Ido Nitzan Hidekel, Dan Raviv

ICML 2026 CATS Workshop — Continual Adaptation at Scale: Towards Sustainable AI. [Paper](#)

Developed a function-space predictor for catastrophic forgetting and showed that forgetting concentrates in a small number of dominant functional modes, motivating targeted regularization methods.

Education

Ph.D., Electrical Engineering (direct track) — Tel Aviv University

2026–present

M.Sc., Electrical Engineering — Tel Aviv University | **GPA 95.5**

2024–2026

B.Sc., Applied Mathematics — Tel Aviv University | **GPA 84**

2020–2024

Technical Skills

Programming & ML. Python, PyTorch, Lightning, Claude Code, LangChain

Methods. Deep learning, foundation-model fine-tuning, representation learning, transfer learning, OOD detection, calibration, statistical evaluation.

Infrastructure. GCP, Vertex AI, PyTorch DDP, Docker, W&B, Linux, Git, distributed experimentation and large-scale data processing.